

The Human Membrane

Packet p. 6 • backup for pp. 11-12

- Today the class becomes a piece of cell membrane.
- You will build a phospholipid bilayer out of people, then test what can cross it and what cannot.
- Same three ideas as the Bubble Lab: the membrane is fluid, it repairs itself, and proteins control what gets through.
- Keep your packet open to page 6. Your labeled diagram is your guide.

Build the bilayer

Packet p. 6 • backup for pp. 11-12

- Two lines, back to back, shoulders touching. Each person is one phospholipid.
- You face OUT: your face is the hydrophilic head, toward the water on each side of the membrane.
- Your backs meet in the middle: your backs are the hydrophobic tails, hiding from the water.
- Special jobs:
 - Channel protein: two students across from each other make an arch with their arms, a doorway through both lines.
 - Cholesterol: one student stands between the two lines and keeps them evenly spaced, not too tight, not too loose.
 - Carbohydrate: one head wears a sticky note “ID” on the outside. It tells other cells who you are.

Concept 1 • The membrane is fluid

Packet p. 6 • backup for pp. 11-12

- On “shuffle,” the heads on the outside line slide one step to the right. Inside line slides left.
- Keep touching shoulders. The membrane never opens up.
- Ask: did the membrane stay a membrane even though every phospholipid moved?
- That is the “fluid” in the fluid mosaic model. The proteins and carbohydrates scattered through it are the “mosaic.”

Concept 2 · The gap closes

Packet p. 6 · backup for pp. 11-12

- One phospholipid steps out of the line.
- Neighbors close the gap right away, without being told.
- Ask: why would the gap close? Heads are attracted to water, tails avoid it, and phospholipids attract each other.
- Same thing the soap film did when a soapy finger pushed through it.

Concept 3 • What gets through?

Packet p. 6 • backup for pp. 11-12

Object we test	Stands for	Predict: straight through, channel only, or blocked?	What happened
Crumpled paper ball	Oxygen or carbon dioxide (small, nonpolar)		
Card marked Na ⁺	A sodium ion (charged)		
Water bottle	Water (small, but polar)		
Textbook	A large protein		

- Rule for the test: an object may slip between two shoulders only if it is small and has no charge. Anything charged must use the arch. Anything too big cannot cross the lines at all.

Check: what the model showed

Packet p. 6 • backup for pp. 11-12

- Paper ball: slips between the shoulders. Small nonpolar molecules cross the bilayer directly.
- Na⁺ card: only through the arch. Charged particles need a channel protein.
- Water bottle: squeezes through slowly, much faster through the arch. Water uses channels called aquaporins.
- Textbook: cannot cross. Large molecules move in vesicles (coming up in transport).
- Selectively permeable: the membrane lets some things cross and keeps others out.
- Where the model is wrong: real phospholipids are not people, and the membrane is about 8 nanometers thick. Name one more way the model is different.